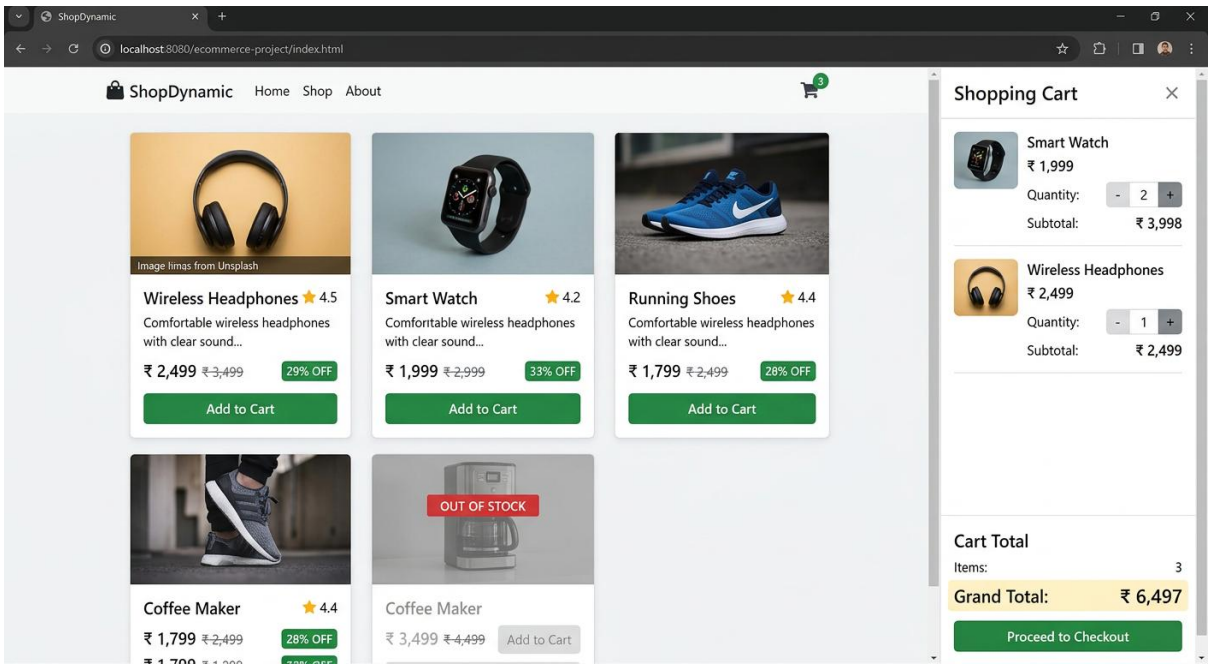
Task: Build a Dynamic E-commerce Product Page Using JavaScript

Create an e-commerce webpage that dynamically displays products as cards using a static array of objects in JavaScript.

Screenshot of expected output:



Requirements

1. Product Display

- Store product details in a static array of objects.
- Each object should contain properties such as product ID, name, price, original price, description, image URL, discount, category, rating, and stock availability.
- Dynamically generate product cards using JavaScript and display them on the webpage.

2. Product Cards

- Each card should display the product image, name, short description, price, original price, and discount.
- Display the "Add to Cart" button at the bottom of each card.

3. Shopping Cart

- When the user clicks the "Add to Cart" button, display a shopping cart section on the right-hand side of the webpage.
- Display the selected product's name, image, price, and quantity in the cart.
- Allow users to add multiple products to the cart.
- If the same product is added more than once, increase its quantity instead of creating duplicate entries.
- Products that are out of stock should not be added to the cart.

4. Cart Total

- Display the quantity of each product and its subtotal.
- Automatically calculate and update the Grand Total whenever a product is added to the cart.
- The grand total should reflect the selling price multiplied by the quantity of each product.

5. Technical Constraints

- Use HTML, CSS, Bootstrap, and vanilla JavaScript.
- Use the provided static array of objects as the product data source.
- Generate product cards and update the cart dynamically using JavaScript DOM manipulation.
- Do not use any additional external JavaScript libraries or frameworks other than Bootstrap.

Expected Layout

- Left-hand side: Product cards arranged in a responsive grid.
- Right-hand side: Shopping cart displaying selected products, quantities, subtotals, and the grand total.

The webpage should have a clean, user-friendly design and work correctly when multiple products are added to the cart.

Product Data

In Point No. 1, use the below array of objects to get the product details:

```
const products = [
  {
    id: 1,
    name: "Wireless Headphones",
    price: 2499,
    originalPrice: 3499,
    discount: 29,
    description: "Comfortable wireless headphones with clear sound and deep bass.",
    image: "https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=500",
    category: "Electronics",
    rating: 4.5,
    inStock: true
  },
  {
    id: 2,
    name: "Smart Watch",
    price: 1999,
    originalPrice: 2999,
    discount: 33,
    description: "A stylish smartwatch for fitness tracking and daily use.",
    image: "https://images.unsplash.com/photo-1523275335684-37898b6baf30?w=500",
    category: "Accessories",
    rating: 4.2,
    inStock: true
  },
  {
    id: 3,
```

```
    name: "Running Shoes",
    price: 1799,
    originalPrice: 2499,
    discount: 28,
    description: "Lightweight running shoes designed for comfort and everyday
workouts.",
    image: "https://images.unsplash.com/photo-1542291026-7eec264c27ff?w=500",
    category: "Footwear",
    rating: 4.4,
    inStock: true
  },
  {
    id: 4,
    name: "Travel Backpack",
    price: 1299,
    originalPrice: 1999,
    discount: 35,
    description: "A spacious backpack suitable for college, office, and travel.",
    image: "https://images.unsplash.com/photo-1553062407-98eeb64c6a62?w=500",
    category: "Bags",
    rating: 4.3,
    inStock: true
  },
  {
    id: 5,
    name: "Digital Camera",
    price: 15999,
    originalPrice: 18999,
    discount: 16,
    description: "Capture memorable moments with a compact digital camera.",
```

```
    image: "https://images.unsplash.com/photo-1516035069371-29a1b244cc32?w=500",
    category: "Electronics",
    rating: 4.6,
    inStock: true
  },
  {
    id: 6,
    name: "Sunglasses",
    price: 899,
    originalPrice: 1499,
    discount: 40,
    description: "Modern sunglasses with a stylish frame for everyday wear.",
    image: "https://images.unsplash.com/photo-1511499767150-a48a237f0083?w=500",
    category: "Fashion",
    rating: 4.1,
    inStock: true
  },
  {
    id: 7,
    name: "Coffee Maker",
    price: 3499,
    originalPrice: 4499,
    discount: 22,
    description: "Prepare fresh coffee at home with this easy-to-use coffee maker.",
    image: "https://images.unsplash.com/photo-1495474472287-4d71bcdd2085?w=500",
    category: "Home Appliances",
    rating: 4.4,
    inStock: false
```

```
  }
];
```